

Seismic Waves and Earth's Internal Structure

An earthquake releases energy and generates seismic waves.

P waves are longitudinal waves. They travel through solids, liquids, and gases.

S waves are transverse waves. They travel only through solids.

Because S waves cannot pass through the liquid outer core, scientists infer that the outer core is liquid.

Arrival times, travel paths, and shadow zones reveal Earth's crust, mantle, outer core, and inner core.